Akhil Indurti

aindurti@gmail.com (703-431-5466)
akhil.cc
github.com/smasher164

SKILLS

Languages C, Go, Rust, C++, Assembly, Java, Python, JS, SML, OCaml, Raku, C#, F#, Racket, Shell, SQL, LaTeX.

Tools HLS, SCTE-35, Widevine DRM, LLVM & Go toolchains, OpenGL, Git, Docker, Kubernetes, Linux, Nix, GraphQL, Android, Adobe Creative Suite, QEMU, Azure, AWS, GCP, DigitalOcean, fly.io, Terraform, ScyllaDB, MongoDB, Concourse, ZFS.

PROJECTS

golang/go Member who contributes patches, triages issues, and reviews code. Works on the Go compiler, standard library, and runtime. Notable contributions: fused-multiply-add, static analysis checks, and playground. **Speaker at GopherCon 2021**: Gave a 45 min talk about "Writing a Static Analyzer for Go Code".

mem A memory allocator for Go.

nas Configuration for Home NAS, running NixOS, ZFS, Tailscale, Home Assistant, Samba, and Plex.

oscons Notes on OS Construction.

mexdown An extensible human-readable plain-text format.

arewesixelyet Site lists support for SIXEL graphics across terminals. Got to front-page of HN.

pw-volume Interface to PipeWire volume controls, before WirePlumber had support. Over 60 active users.

gflat An ML-like programming language with rank-n polymorphism, typeclasses, algebraic data types, green threads, and data-race freedom.

burrito Build system and package manager on an ML-like language with lexical effect handlers.

tapl Typed lambda calculus solutions for learning type theory from *Types and Programming Languages*. **Participant at OPLSS 2021**: Participated in the Oregon Programming Languages Summer School to study interesting topics in PL Theory and Formal Verification.

EXPERIENCE

Engineer III, Software Process Engineering @ Los Angeles, CA *Samsung Ads*, May, 2026 - Present
Maintain the Origin Manifest Manipulator, Pipeline, and Sessions services that deliver Samsung TV Plus content and handle server-side ad insertion (SSAI).

- Migrated the legacy Origin service into Pipeline, and rolled out Transmit as an alternative SSAI provider to AWS Elemental MediaTailor.
- Descaled the Sessions service to cut infrastructure costs, and built dashboards and telemetry for stream and ad-insertion health.

Software Engineer II on Azure @ Redmond, WA *Microsoft*, March, 2023 - Nov, 2025
Service Fabric team: System Services (Failover Manager, Transport, Federation) and Infrastructure Services.

- Worked on routing, failover, networking, and VM upgrades for the distributed systems platform that underpins much of Azure.
- On call for live-site incidents, driving mitigations and RCAs.

Software Engineer II on .NET @ Redmond, WA *Microsoft*, May, 2022 - March, 2023

- **APICompat**: Worked on the public rollout for a tool that detects binary breaking changes between two .NET assemblies.
- **GenAPI**: Generates C# source from an assembly. The generated source contains stub declarations for types and members, where declarations preserve the API contract of the original source. Fixed Roslyn bugs for assembly metadata.
- **Doc source porting tool**: Spearheaded internal effort to port externally maintained XML docs into inline docs for the .NET repos.

Software Engineer II @ Mount Laurel, NJ *Comcast*, May, 2020 - April, 2022
Working on Comcast's Next Generation Access Network (NGAN) to build Genome, a poller for millions of network devices across the country.

- Implemented an SNMP connection pool that significantly sped up the long tail of our polling cycle, and cut AWS costs.
- Led a testing & deployment strategy to discover test devices and monitor health statuses.
- Detected incompatible changes to GraphQL and Avro schemas.
- Automated datalake topic generation and IAM roles via Terraform.
- Assisted in MongoDB to ScyllaDB migration, and multi-region setup.
- Improved and hired talent to improve code quality and CI infrastructure.

Research Assistant @ Charlottesville, VA *ShiftLab*, Jan, 2020 - May, 2020
Worked under Professor Samira Khan to find crash-consistency bugs via structured fuzzing, as well add support for persistent memory to the Go programming language. Contributed to:

- Cross-Failure Bug Detection in Persistent Memory Programs
- PMFuzz: test case generation for persistent memory programs

Teaching Assistant @ Charlottesville, VA *University of Virginia*, Spring 2018 - Spring 2019

- **Theory of Computation**: Course covers automata theory, complexity theory, and problem-solving techniques. I assist in grading, holding office hours, and supervising problem-solving sessions. I developed a new Turing Machine Simulator assignment.
- **Program Data and Representation**: Course covers C++, data structures, and assembly. I assist in grading, holding office hours, and running lab. I developed a new Tree lab that introduces AVL trees.

Research @ Charlottesville, VA *Center for Automata Processing*, Fall 2017 - Spring 2018
I worked to explore the application and implementation of state machines. Under my advisors Professor Kevin Skadron and Jack Wadden, *my research* focused on developing and implementing an NFA reduction algorithm to perform prefix and suffix merging in VASim.

Android SDK @ Palo Alto, CA *TRNQL*, January - August 2015
Worked to enable easy integration of contextual information into an application. Created example applications for clients to study, and led meetups to introduce the community to these SDK features.

Software Engineering Intern @ Milpitas, CA *FireEye*, Summer 2014
Implemented a web application for the management and education on the company's security products.

- Wrote a web application to educate, display, and manage the new intrusion prevention system.
- Developed a real-world understanding of network security in the context of large organizations.

EDUCATION

University of Virginia (B.S. in Computer Science)
Coursework: Operating Systems, Advanced Computer Architecture, Graduate Compilers, Machine Learning, Computer Graphics, Capstone Research, Algorithms, Theory of Computation, Program Data and Representation